

WEEK 12

Protein structures: prediction, confidence and function

What does a confident structure still fail to prove?

By the end, you should be able to...

Distinguish sequence, structure and function.

Interpret pLDDT and PAE without turning them into binding probabilities.

Propose an experimental test for a structure-based hypothesis.

Reconnect with the biology

A protein is an amino-acid chain; its structure and interactions depend on sequence and biological context.

Think silently -> explain to a neighbour -> check the measurement.

Structure is one layer of evidence

01

A sequence specifies an amino-acid chain.

>

02

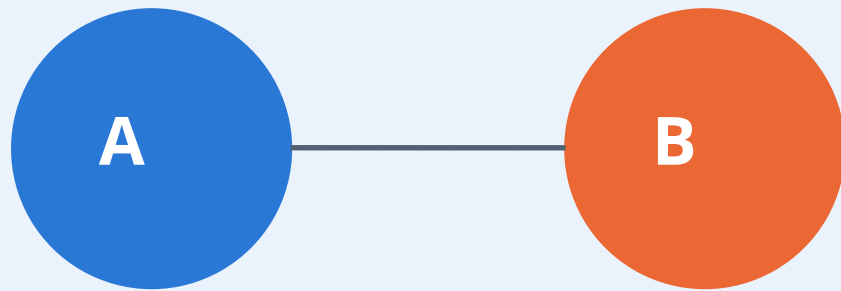
An experimental structure and a computational prediction have different provenance.

>

03

Neither a picture nor a confidence score is itself a functional assay.

Proteins can have domains and flexible regions



Schematic domains + flexible linker
Not a molecular structure

A domain is a structurally or functionally meaningful region.

Flexible linkers may allow different arrangements.

A single structure does not capture every state or partner.

pLDDT is local model confidence

<50 very low

50-70 low

70-90 confident

>90 very high

Guidance bands, not functional certainty.

AlphaFold pLDDT is a per-residue score on a 0-100 scale.

Higher values generally indicate greater confidence in local structure.

It is not the percent probability that a protein binds its target.

PAE asks about relative placement

Synthetic block PAE (Angstrom)

Aligned on	Domain A	Domain B
Domain A	2	18
Domain B	20	3

Low diagonal; high off-diagonal
Domain orientation remains uncertain

Predicted aligned error estimates positional error relative to an alignment reference.

Low within-domain PAE and high between-domain PAE can indicate uncertain domain arrangement.

Check units and the matrix axes.

Low confidence has several explanations

It may reflect disorder, flexibility, missing context or a prediction error.

Do not automatically label every low-score region disordered.

Use complementary evidence and state alternatives.

A real database card, not a new experiment

98.06

Displayed global pLDDT; record
snapshot

AF_AFP69905F1 is a computed
hemoglobin alpha model.

The RCSB record displayed 142 modeled
residues and global pLDDT 98.06 when
checked.

The record explicitly distinguishes this
model from experimental validation.

An interaction score is not an affinity measurement

Model confidence

Measured function

Model confidence, geometric plausibility and binding affinity are different quantities.

A predicted complex can motivate testing, not replace it.

Binding and specificity need appropriate positive and negative experimental controls.

The Structure Report Card

Sequence / model

Record sequence / accession and the model source.

Local confidence

Describe local confidence and uncertain relative placement.

Relative placement

Claim + validation

Name the proposed use, a limitation and a validation experiment.

Worked example: two confident domains

A toy protein has two domains with pLDDT near 90, a linker near 45 and between-domain PAE near 20 angstroms.

Predict the result and one limitation before calculating.

Work through the reasoning

Local domain geometry may be more reliable than the linker.

Relative domain orientation is uncertain in this example.

Do not design an interface hypothesis that assumes one exact orientation without further evidence.

Pause: make a defensible claim

A model has pLDDT 95. A report says there is a 95% chance that the protein binds a proposed target. What is wrong?

First answer alone. Then compare the evidence, not just the conclusion.

A defensible response

pLDDT is not a binding probability. The claim confuses local structural confidence with interaction evidence.

Today's practical

W12 studio: inspect supplied synthetic confidence values and the real hemoglobin database evidence card; write a bounded structure interpretation.

Minutes	Action
0-10	Predict + set up
10-30	Run + inspect + check
30-40	Explain + compare
40-45	Save + submit / exit

Five terms to use precisely

1 domain

2 pLDDT

3 PAE

4 confidence

5 affinity

Exit ticket

What did we actually measure or calculate?

What is one assumption or limitation of that result?

What check would you perform before sharing it?

Go further only after the core is clear

Optional: inspect a predicted complex with interface metrics; no docking or molecular dynamics is required.

Not required for the core grade. No advanced model training is expected.

Read the evidence, not only the slide

- S01 Alberts et al. Molecular Biology of the Cell, 4th ed. Foundational cell biology chapters.
- S16 EMBL-EBI AlphaFold training: evaluating predicted structures using confidence scores.
- S17 Jumper et al. (2021). Highly accurate protein structure prediction with AlphaFold.
- S24 RCSB PDB computed model AF_AFP69905F1 / AlphaFold DB P69905.